

CS 310: Assignment 3

Question 1.

For the languages below, either construct an NFA that accepts them or prove that the language is not regular.

- (a) $\{w \in \{0,1\}^* \mid w \text{ contains an even number of 0s, or exactly two 1s}\}$
- (b) $(L(ab^*a)) \cap L(a^*b^*a)$
- (c) $L = \{www \mid w \in \{a,b\}^+\}$
- (d) $L = \{w \in \{a,b\}^* \mid \#a(w) > \#b(w)\}$

Assignment 3

1 a)

b)

c) given the pump number n

word $\Rightarrow a^n b^n a^n b^n a^n b^n$

for any x, y s.t. $|xy| \leq n$
 $y \neq \epsilon$

$\Rightarrow xy$ is ~~just~~ consists of only a 's
 let $y = a^i$ $x = a^{n-i}$

$\Rightarrow$ pump y once

$xy^2z = a^{n+i} b^n a^n b^n a^n b^n$ ~~not~~

not in L $\therefore$ not regular

d) $L = \{w \in \{a,b\}^* \mid w \text{ has more } a\text{'s than } b\text{'s}\}$

for pump number n
 $w \Rightarrow b^n a^{n+1}$

for any prefix xy s.t. $|xy| \leq n$
 $y \neq \epsilon$

$\Rightarrow xy$ consists of only b 's
 and y is non empty
 $x = b^i$ $y = b^j$

$\Rightarrow$ pump y once

$xy^2z = b^{n+j+1} a^{n+1}$ ~~not in L~~ as $j \geq 1$

Solution: 1(a,b,c,d): 23B1043

(e) $L = \{w \in \{a, b\}^* \mid \#a(w) \neq \#b(w)\}$

(e) $L = \{w \in \{a, b\}^* \mid \#a(w) \neq \#b(w)\}$
 Consider the complement of $L = \{w \in \{a, b\}^* \mid \#a(w) = \#b(w)\} = E$

$S = a^p b^p \in E$
 $S = xyz$

$aa \dots aa \dots aab^p$
 $\quad \quad \quad x \quad y$

$|y| > 0$

When you pump: S has more a 's than b 's $\notin E$
 so E is not regular

$\therefore L$ is complement of E , L cannot be regular

Solution: 1(e): 23B0936

(f) $L = \{a^i b^j a^k \mid i + j = k\}$

(f) $L = \{a^i b^j a^k \mid k \neq i + j\}$

$E = \{a^i b^j a^k \mid k = i + j\} = S$

Assume S is regular, p : pumping length

$s = a^p b^p a^{2p} \in S$

Pump with $n=0$

$xy^0z = a^{p-t} b^p a^{2p}$
 $\notin S$

$a \dots ab \dots ba \dots a$
 $\quad \quad \quad x \quad y \quad z$

$\leq p \quad 1 \leq t \leq p$

$|y| = t$

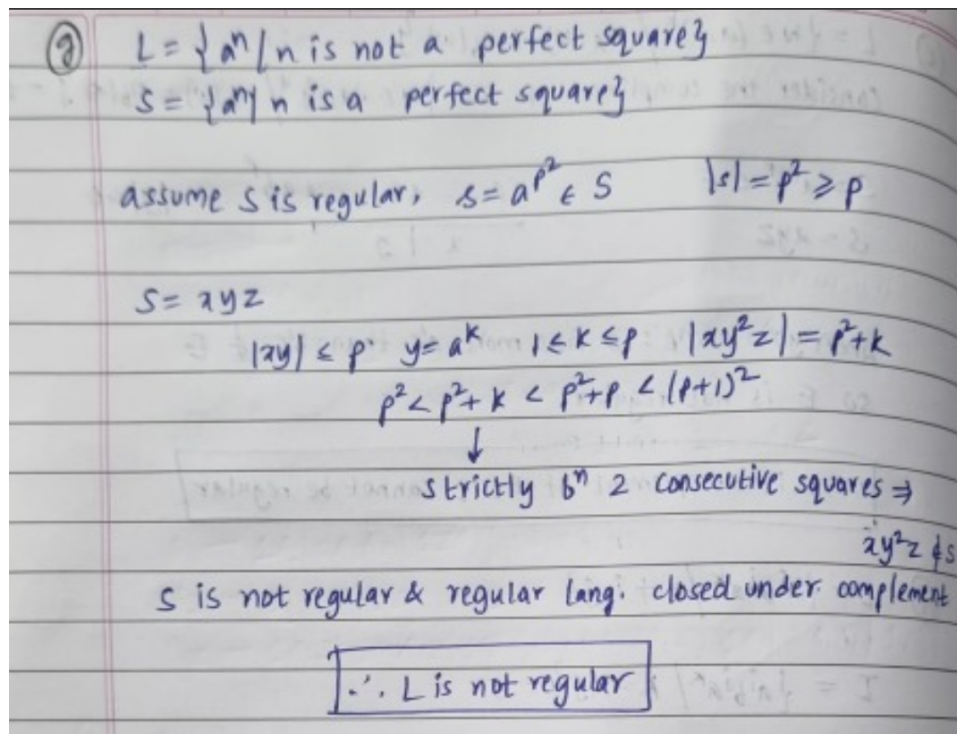
$\therefore S$ is not regular

$\downarrow$
 closed under complement

$\therefore L$ is not regular

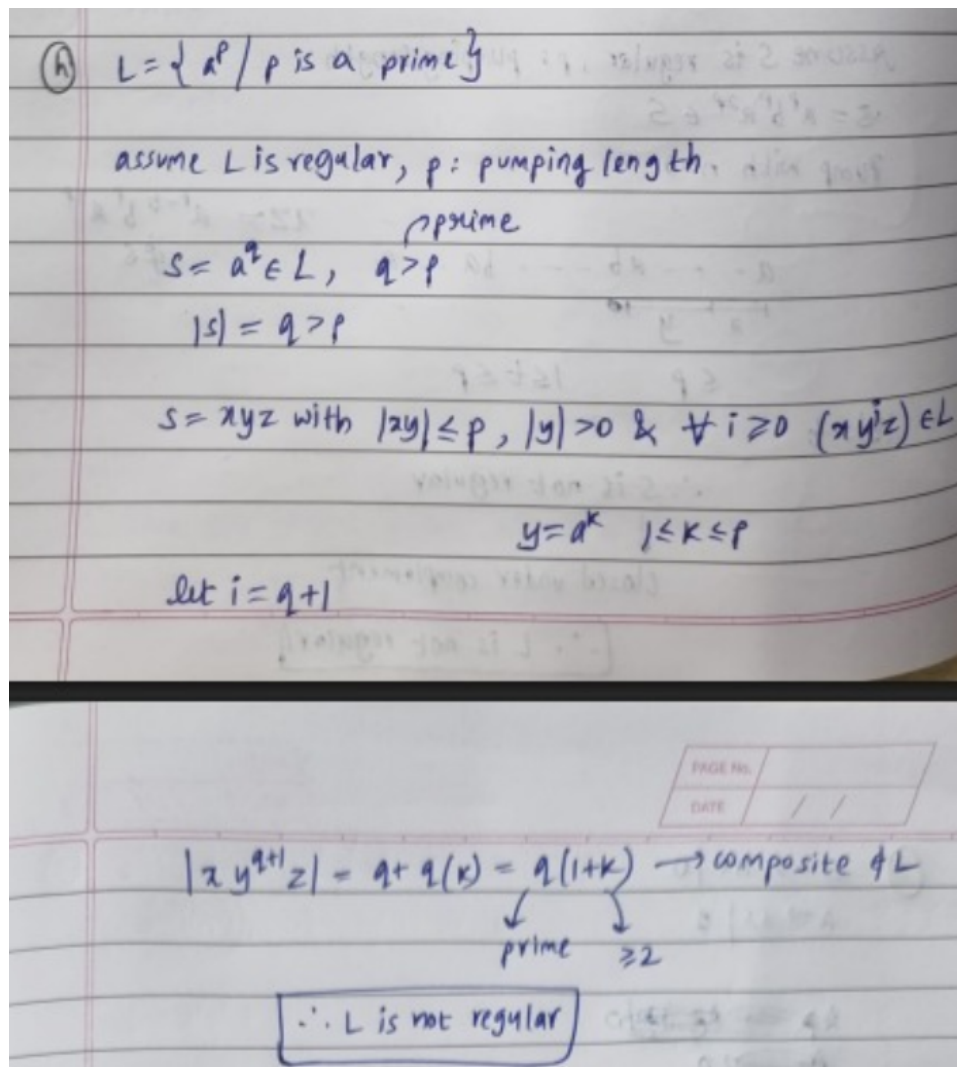
Solution: 1(f): 23B0936

(g) $L = \{a^n \mid n \text{ is not a perfect square}\}$



Solution: 1(g): 23B0936

(h) $L = \{a^p \mid p \text{ is a prime number}\}$



Solution: 1(h): 23B0936

Question 2.

Convert the following CFG into an equivalent CFG in Chomsky Normal Form:

$$S \rightarrow PSP \mid P \mid \epsilon$$

$$P \rightarrow 00 \mid \epsilon$$

$S_0 \rightarrow PQ \mid SP \mid PS \mid xx \mid \epsilon$
$S \rightarrow PQ \mid SP \mid PS \mid xx$
$P \rightarrow xx$
$Q \rightarrow SP$
$x \rightarrow \emptyset$

Solution 2: 23B0989

Question 3.

Convert to Greibach Normal Form:

$$S \rightarrow AA \mid 0$$

$$A \rightarrow SS \mid 1$$

$S \rightarrow AA \mid 0$
 $A \rightarrow SS \mid 1$
 ↳ it is in CNF

$S \rightarrow AA \mid 0$
 $A \rightarrow AAS \mid OS \mid 1$
 ↓
 $S \rightarrow AA \mid 0$
 $A \rightarrow OSZ \mid 1Z \mid OS \mid 1$
 $Z \rightarrow AS \mid ASZ$

GNF:

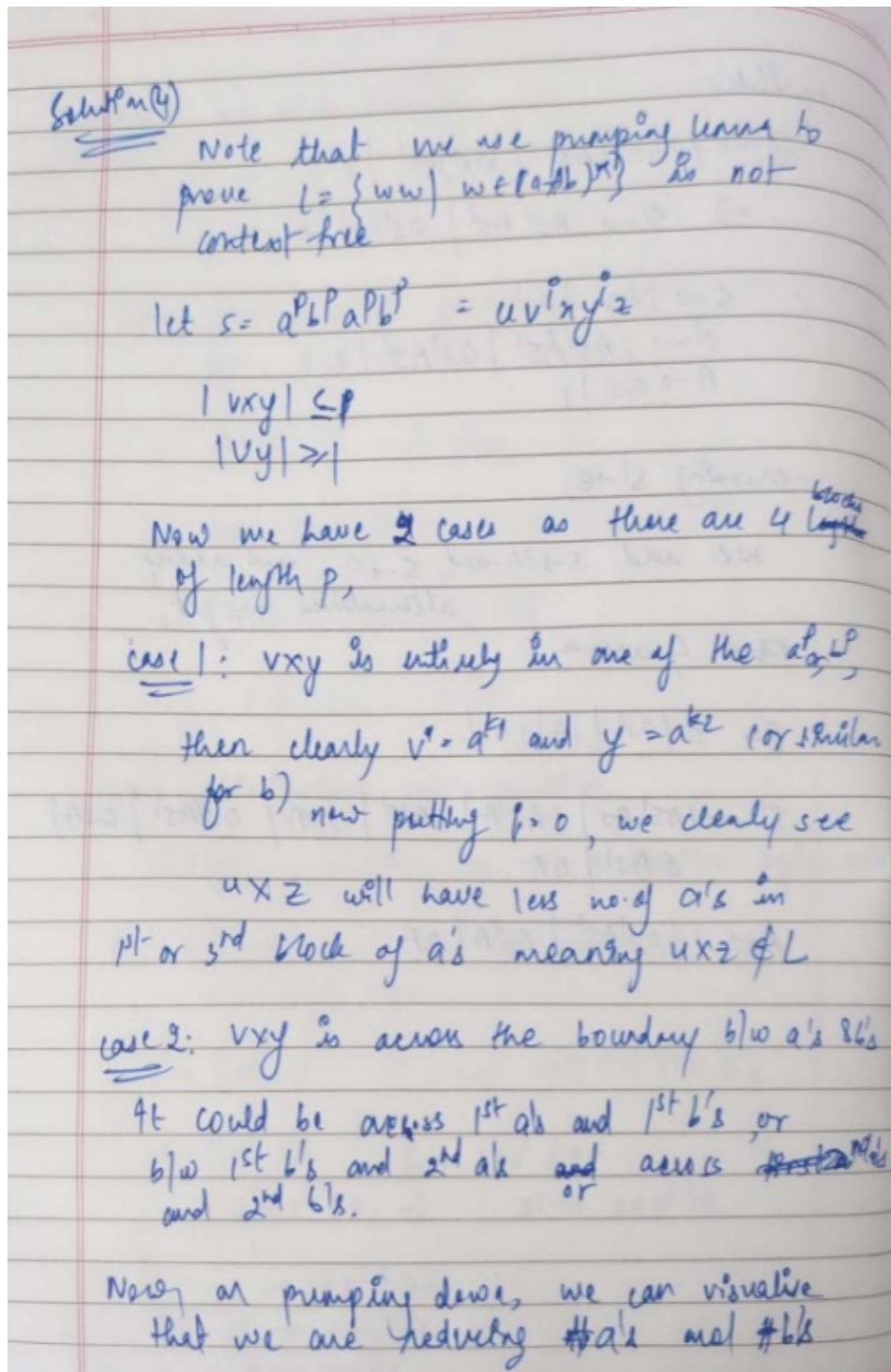
$S \rightarrow OSZA \mid 1ZA \mid OSA \mid 1A \mid 0$
 $A \rightarrow OSZ \mid 1Z \mid OS \mid 1$
 $Z \rightarrow OSZS \mid 1ZS \mid OSS \mid 1S \mid OSZSZ$
 $\quad \mid 1ZSZ \mid OSSZ \mid 1SZ$

Solution 3: 23B0943

Question 4.

Show that $L = \{ww \mid w \in (a+b)^*\}$ is not context-free.

Is the complement of L (strings not of the form ww) context-free? Prove or disprove.



but asymmetrically i.e. say I reduce the no. of a's b/w 1st & 2nd block, in the

new format 1st half is no longer equal to second half & same can be argued for all 3 cases,

hence in any word of L $|u| \neq |v|$

hence we can conclude L is not context-free

⇒ For complement of L , we can give explicit grammar to show it is context-free

$S \rightarrow A \mid B \mid AB \mid BA$

$A \rightarrow a \mid aAa \mid aAb \mid bAb \mid bAa$

$B \rightarrow b \mid bBb \mid bBa \mid aBb \mid aBa$

Explanation: A intuitively is all odd length words with middle letter a and B is all odd length words with middle letter b

So A, B i.e. all odd length strings directly belong to T , for even length AB & BA have atleast one position in 1st half that differs from one in 2nd half

hence T is context-free, proven by giving explicit grammar

Solution 4: 23B0942

Question 5.

Given a language L we define its reversal as

$$L^R = \{w^R \mid w \in L\}.$$

Prove (informally) or disprove that the set of context-free languages is closed under the reversal operation.

⑤ $L^R = \{w, w^R \in L\}$

CFG - $G = (\Sigma, V, S, P)$

$G' = (\Sigma, V, S, P')$ - reversing RHS of every prod in P .

If $A \rightarrow X_1 X_2 \dots X_n$ is a rule in P , then

$A \rightarrow X_n X_{n-1} \dots X_1$ in P'

If $A \xRightarrow{*}_G \alpha$, then reverse $A \xRightarrow{*}_{G'} \alpha^R$

$\therefore S \xRightarrow{*}_G w \text{ for } w \in L(G) \Rightarrow S \xRightarrow{*}_{G'} w^R \text{ for } w^R \in L(G)^R$

$\therefore L(G) = L(G')^R$

L^R is generated by CFG G'

CFG closed under reversal operation.

Solution 5: 23b1030